

Xingdong Zuo

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Google Scholar

Experience

May 2021 – present **Research Engineer**, Naver Corp – South Korea

- Built auto-bidding for display ads with offline RL (customized IQL) trained on large-scale logged interaction data, achieving a double-digit improvement in ads efficiency; presented at Naver DAN 2025.
- Improved CVR and ROAS of NAVER Dynamic AD with advanced retrieval and RL models.
- Developed RL recommendation models for LINE Wallet, optimizing long-term user engagement.
- More: prototyped RLHF for user foundation models; integrated offline RL into Smart Message ranking; improved embedding and reranker models for Brand Chatbot; built the QueryLake ETL library and KPI reporting pipelines.

May 2017 – Nov 2017 **Master Thesis**, Bosch Center for Artificial Intelligence – Germany

Sample-efficient model-based policy search via Bayesian Neural Network dynamics models.

- Improved sample efficiency of model-based RL on continuous control from ~10 to ~6 trials.

Education

2015 – 2019 **University of Freiburg**, M.Sc. in Computer Science – Freiburg, Germany
Focus: Machine Learning and Reinforcement Learning

2010 – 2015 **Linnaeus University**, B.Sc. in Applied Mathematics – Vaxjo, Sweden
Minor: Computer Science

Projects

2026 – present **Drift-VLA — Direct One-Step VLA Action Generation**

Research on VLA inference via one-step drifting, replacing iterative flow matching.

- Proposed **action-dimension drifting**.
- Faster decoding by 3–4×: 85.6 vs 259.8 ms/chunk for Pi0.5.
- Real LeKiwi task: 9/10 rollouts. On LIBERO, Pi0.5-Drift reaches 97.4% at 1 NFE against 96.9% for flow matching at 10 NFE.

2026 – present **LeRobot CAD Studio (author)**

Browser studio to design 3D-printed parts for LeRobot robots with an AI coding agent, check them, and test them in MuJoCo. Live demo.

- Agent workflow: turns a part request into a prompt and context pack; the returned STL becomes a new part option.

- Automatic checks: printability, fit and arm clearance, joint limits, camera view.
- In-browser MuJoCo (WASM) simulation of LeKiwi + PincOpen with a lab sample-transport task; exports STL and MJCF.

2026 – present **Mobile Manipulation with LeKiwi and PincOpen — HF Blog**

- Integrated the PincOpen gripper into LeKiwi: published a detailed build tutorial, and diagnosed an arm-extension voltage-drop failure.

2025 – 2025 **LeRobot Worldwide Hackathon 2025 — Hugging Face**

Led a 5-person team over 2 days to complete a language-instructed manipulation task using SmolVLA on real hardware.

- Owned team coordination, task decomposition.

2026 – present **lekiwi-tui (author)**

Terminal control center for the LeKiwi robot workflow, built on LeRobot. Also lerobokinson (2025), a GUI for PD gain tuning of Dynamixel servos.

Publications

2026 **Constrained Auto-Bidding via Generative Response Modeling**
E Yang, X Zuo, K M Kim
dl.acm.org/doi/10.1145/3770855.3817847 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD))

2025 **Method and System to Provide Information Data with Minimal Infrastructure Resources**
X Zuo, et al.
patents.google.com/patent/KR20250124713A/en (Korean Patent Publication No. 10-2025-0124713)

2024 **HyperCLOVA X Technical Report**
HyperCLOVA X Team (incl. X Zuo)
arxiv.org/abs/2404.01954 (arXiv preprint)

2023 **Direct preference-based policy optimization without reward modeling**
G An, J Lee, X Zuo, N Kosaka, K M Kim, H O Song
arxiv.org/abs/2301.12842 (Advances in Neural Information Processing Systems 36)

2014 **Numerical simulation of Asano-Khrennikov-Ohya quantum-like decision making model**
X Zuo
(NeuroQuantology 12 (4))

Skills

Robot Learning: Vision-Language-Action models (SmolVLA, PI05, ACT), LeRobot, SO-101 & LeKiwi, PincOpen gripper, Feetech/Dynamixel servo

Reinforcement Learning: Offline RL (IQL, CQL), PPO, model-based RL (PILCO)

Simulation & Evaluation: MuJoCo (RL policy training), LIBERO benchmark, OpenAI Gym / Gymnasium

ML Engineering: PyTorch, multi-GPU distributed training, Kubeflow, Docker, Polars, Trino SQL, Qdrant, Linux, Git, Claude Code

Languages: English (fluent), Chinese (native)

Work Authorization: E-7 visa (Korea)

Open Source

2025 – present **LeRobot Contributor (11 PRs, #11 contributor in 2026) — Hugging Face**

- Fixed SmolVLA VLM layer freezing under `train_expert_only=False` (#4019)
- Batched video encoding for faster dataset recording (#1390)
- LeKiwi observation streaming over ZMQ multipart, 25% less bandwidth (#4088); LeKiwi support in the replay and rollout/eval CLIs (#3739, #3742)
- LIBERO reproducibility docs for the `lerobot/libero` dataset and pi05 quick-start (#4185, #4186)
- Open PRs — SmolVLA inference optimization: RoPE cache precompute (#4372), SDPA attention with KV/mask reuse (#4181); trackio tracking (#4320)

2025 – present **rlbidder (author)**

Modular reinforcement learning auto-bidding library for research and production.

2018 – 2020 **OpenAI Gym Contributor (42 PRs, top #5 contributor, Nov. 2019)**

- Implemented 12 of the 14 environment Wrapper classes
- Refactored spaces (`flatdim`, `flatten`, `unflatten`) and vectorized-environment infrastructure

2018 – 2020 **PyTorch Contributor (31 PRs)**

- `TanhTransform` for constrained distributions (#19785, #40160)
- DataLoader `drop_last` for non-divisible dataset sizes (#1133)
- Also `parameters_to_vector` (#2795), `ModuleList.insert` (#11664), and a numerical-stability fix in `SigmoidTransform` (#19802)

2018 – 2021 **lagom (author)**

PyTorch library for rapid prototyping of reinforcement learning algorithms (378★).

2017 – 2020 **Other libraries (author)**

mazelab (271★), VIN (223★), dm2gym (87★), DeepPILCO (54★).